

Lesson 14: Quadratic Functions & the Discriminant — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Solve $x^2 - 7x + 12 = 0$.
2. Solve $x^2 - 9 = 0$.
3. Solve $x^2 + 5x + 6 = 0$.
4. Factorise $x^2 + 2x - 15$.
5. Write down the y-intercept of $y = x^2 - 3x + 7$.
6. Solve $x^2 - 4x = 0$.

Section 2 — Core practice

7. Solve $x^2 - 2x - 8 = 0$.
8. Find the discriminant of $x^2 + 3x + 1$.
9. Find the discriminant of $x^2 - 4x + 4$.
10. How many real roots has $x^2 + x + 5 = 0$?
11. Solve $2x^2 - 5x + 2 = 0$.

12. Solve $x^2 + 6x + 9 = 0$.

Section 3 — Challenge & exam-style

13. $x^2 + kx + 16 = 0$ has equal roots ($k > 0$). Find k .

14. Find the values of k for which $x^2 + kx + 9 = 0$ has two distinct real roots.

15. Solve $x^2 - 6x + 4 = 0$ with the quadratic formula (3 s.f.).

16. For what value of c does $x^2 - 8x + c = 0$ have a repeated root?